

Arnab Jena

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Education

Manipal Institute of Technology, Bengaluru

Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence); **CGPA: 7.54/10.0** Aug 2023 – Jul 2027

Indian Institute of Technology Madras

Bachelor of Science in Data Science and Applications; **CGPA: 7.14/10.0**

Bengaluru, India

Chennai, India

Jan 2024 – Present

Experience

Software Developer Intern

Jan 2026 – Present

Center for Data for Public Good, Indian Institute of Science

Bengaluru, India

- Contributed to the WASP traffic-simulation platform by studying and mapping Eclipse SUMO/netedit features, tracing relevant C++ components, analysing Network, Demand and Traffic Light System workflows, and documenting their porting requirements for the unified Qt6-based interface.
- Worked on an India-focused Driver Behaviour Model by reconstructing the HSFM calibration workflow in Python, reproducing historical calibration results, running multi-seed validation, analysing trajectory-based longitudinal interactions, and defining the numerical reference for later C++/SUMO integration.

Research Intern

May 2025 – Jul 2025

IPSA Lab, Indian Institute of Technology Ropar

Rupnagar, India

- Analyzed temporal NDVI patterns from Sentinel-2 and Landsat-8 imagery and built crop-detection workflows in Google Earth Engine, including preprocessing, time-series visualization, classification, and validation.
- Developed annotation and evaluation pipelines for YOLO-based object detection on X-ray, CT, and MRI datasets; co-authored a CVIP 2025 paper on automated NDVI rice detection using satellite imagery.

Frontend Developer Intern

Aug 2024 – Jul 2025

InternLoom

Bengaluru, India

- Built responsive product interfaces using Next.js, Tailwind CSS, Firebase, and MongoDB for an internship discovery platform with search, filters, authentication, and user-facing dashboards.
- Improved navigation, performance, and visual consistency across core user flows through iterative testing, component refactoring, and design updates.

Projects

Pixlate | Next.js, OCR, Translation, Image Processing

[Live](#)

- Built a generative art system that combines HSL-based color selection with KD-tree spatial search to create organic, coherent compositions while keeping placement efficient as the canvas grows.
- Designed a **4-stage pipeline** for text detection, translation, inpainting, and review-ready rendering to simplify multi-market creative adaptation.

Colora | JavaScript, HTML, CSS, Canvas

[Live](#)

- Created a calm, pastel-styled PDF annotation toolkit, freehand drawing, text, shapes, notes, signatures, page controls, zoom, and export support.

In.decoded | SvelteKit, D3.js, Scrollama, GeoJSON

[Live](#)

- Built an experimental long-form data story that makes complex public data more interpretable through narrative and interaction.
- Visualized India's delimitation debate with **7+ interactive maps and charts** covering population, representation, fertility, GDP, tax transfers, and literacy.
- Implemented scroll-driven cartograms, state-level comparisons, and animated transitions using D3.js, Scrollama, and custom GeoJSON projections.

VOXI | TypeScript, Three.js, Rust, WebSockets, LiveKit

[Live](#)

- Built an interactive 3D voxel creation platform with solo and real-time multiplayer rooms, synchronized building, shareable room codes, and integrated voice communication.

Visionics | Next.js, YOLO, Firebase, Data Visualization

[GitHub](#)

- Developed an AI-powered system combining a digital twin dashboard and YOLO-based PPE detection for real-time safety monitoring. Enabled zone-wise risk analysis, live alerts, and compliance tracking using interactive dashboards and Firebase.
- Secured **4th place among 150+ teams** in a national hackathon organized by Hitachi Research.

Technical Skills

Languages: C, Python, Java, JavaScript, TypeScript, Go, SQL

Frameworks: React, Next.js, SvelteKit, Tailwind CSS, Node.js, Express, Firebase, MongoDB

Data Visualization and BI: Tableau, Power BI, D3.js, p5.js, SQL, QGIS, Scrollama

Data and ML: pandas, NumPy, scikit-learn, TensorFlow, OpenCV, YOLO, Google Earth Engine

Tools: Git, Vercel, Jupyter, Google Colab, Figma

Leadership and Achievements

- Co-authored a research paper titled “Evaluating Social Biases in IndicBERT, mBERT, and Phi-3 Using the Indic-Bias Benchmark for the Indian Context”, accepted at the 6th International Conference on Innovations in Computational Intelligence and Computer Vision (ICICV 2025) at Binus University, Jakarta, Indonesia.
- ICICV 2025 presentation certificate: [View certificate on Google Drive](#)**
- Co-authored a CVIP 2025 research paper on temporal NDVI analysis and automated rice crop detection using Sentinel-2, Landsat-8, and Google Earth Engine.
- Quiz Head, Minamansa Quiz Society:** Organized **3+ campus quizzes** and represented MIT Bengaluru at Tata Crucible, Quiz on the Beach at TAPMI, UTPT at RVCE Bengaluru, and IISc quiz events.
- National hackathon:** Secured **4th place among 150+ teams** at Hitachi Research for Visionics, an AI safety digital twin with YOLO-based PPE monitoring.